Properties of Acids & Bases

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Properties of acids

- Acids have pH values of below 7, have a sour taste (when edible) and are corrosive
- Acids are substances that can neutralise a base, forming a salt and water
- When **acids** are added to water, they form positively charged **hydrogen ions** (H^+)
- The presence of H^+ ions is what makes a solution acidic

Example: Hydrochloric acid

$$HCl\,(aq) \rightarrow H^+\,(aq) + Cl^-\,(aq)$$

Acids and metals

- Only metals **above hydrogen** in the reactivity series will react with dilute acids.
- When acids react with metals they form a salt and hydrogen gas:

acid + metal → salt + hydrogen

- The name of the salt is related to the name of the acid used, as it depends on the **anion** within the acid

Examples of the names of salts from specific acids and metals

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Hydrochloric acid	Magnesium chloride and hydrogen	$Mg + 2HCl \rightarrow MgCl_2 + H_2$
Sulfuric acid	Magnesium sulfate and hydrogen	$Mg + H_2SO_4 \rightarrow MgSO_4 + H_2$
Nitric acid	Magnesium nitrate and hydrogen	$Mg + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2$

Help me understand

Acids with bases

- Metal **oxides** and metal **hydroxides** (**alkalis**) can act as **bases**
- When they react with acid, a **neutralisation** reaction occurs
- In all acid-base neutralisation reactions, **salt** and **water** are produced

acid + base → salt + water

Examples of reactions between acids and bases

Acid	Name of products	Equation for reaction
Hydrochloric acid	Magnesium chloride and water	$Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$
Sulfuric acid	Magnesium sulfate and water	$MgO + H_2SO_4 \rightarrow MgSO_4 + H_2O$
Nitric acid	Magnesium nitrate and water	$Mg(OH)_2 + 2HNO_3 \rightarrow Mg(NO_3)_2 + 2H_2O$

Acids with metal carbonates

- Acids will react with metal carbonates to form the corresponding metal salt, carbon dioxide and water:

acid + metal carbonate → salt + carbon dioxide + water

Examples of reactions between acids and carbonates

Acid	Name of products	Equation for reaction
Hydrochloric acid	Magnesium chloride, carbon dioxide and water	$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$

Nitric acid	Magnesium nitrate, carbon dioxide and water	$MgCO_3 + 2HNO_3 \rightarrow Mg(NO_3)_2 + CO_2 + H_2O$

What are indicators?

- Two colour indicators are used to distinguish between acids and alkalis
- Many plants contain substances that can act as indicators and the most common one is **litmus** which is extracted from lichens
- Synthetic indicators are organic compounds that are sensitive to changes in acidity and appear different colours in acids and alkalis
- **Thymolphthalein** and **methyl orange** are synthetic indicators frequently used in acid-alkali titrations

Two Colour Indicators Table

Indicator	Colour in acid	Colour in alkali
Litmus	red	blue
Thymolphthalein	colourless	blue
Methyl orange	red	yellow

What are synthetic indicators?

- Synthetic indicators are used to show the endpoint in titrations as they have a very sharp change of colour when an acid has been neutralised by alkali and vice-versa
- Litmus is not suitable for titrations as the colour change is not sharp and it goes through a purple transition colour in neutral solutions making it difficult to determine an endpoint
- Litmus is very useful as an indicator paper and comes in red and blue versions, for dipping into solutions or testing gases

Properties of bases & alkalis

- Bases have pH values of above 7
- A base which is water-soluble is referred to as an **alkali**
- In basic (alkaline) conditions red litmus paper turns **blue**, methyl orange indicator turns **yellow** and thymolphthalein indicator turns **blue**
- Bases are substances which can neutralise an acid, forming a salt and water
- Bases are usually **oxides** or **hydroxides** of metals

solution an alkali

Example: Sodium hydroxide

$$NaOH\,(s) \rightarrow Na^{+}\,(aq) + OH^{-}\,(aq)$$

Bases and acids

- When bases react with an acid, a neutralisation reaction occurs
- Acids and bases react together in a neutralisation reaction and produce a salt and water:

Acid + base → salt + water

Examples of reaction between bases and acids

Acid	Name of products	Equation for reaction
Hydrochloric acid	Magnesium chloride and water	$Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$
Sulfuric acid	Magnesium sulfate and water	$MgO + H_2SO_4 \rightarrow MgSO_4 + H_2O$
Nitric acid	Magnesium nitrate and water	$Mg(OH)_2 + 2HNO_3 \rightarrow Mg(NO_3)_2 + 2H_2O$

Alkalis and ammonium salts

- Ammonium salts undergo **decomposition** when warmed with an alkali
- Even though ammonia is itself a weak base, it is very **volatile** and can easily be displaced from the salt by another alkali
- A salt, water and ammonia are produced
  - For example:

$$\mathbf{NH_4Cl + NaOH \rightarrow NaCl + H_2O + NH_3}$$

- This reaction is used as a chemical test to confirm the presence of the ammonium ion (NH_4^+)
- Alkali is added to the substance with gentle warming followed by the test for ammonia gas using damp red litmus paper
- The damp litmus paper will turn from **red** to **blue** if ammonia is present

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